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1 Exercise 09

1.1 Exercise 1

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Consider the fib_recursive2(n: u32) function in fibonacci.rs file given by Exercise 4 (see bel

Let PA1 be the parallel version of fib_recursive2(n: u32) function, where the recursive calls

Please answer the questions below and use big-O notation.
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[ ]:
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fn fib_recursive2(n: u32) -> u64 {
if n <= 2 { 1 } else { fib_recursive2(n - 1) + fib_recursive2(n - 2) } }
```

1.1.1 (a) What is the time complexity of PA1 if it is run sequentially, i.e., on a single processor?

Sequentially :

$$T(n) = T(n-1) + T(n-2) + O(1)$$
$$T(n) = O(2^n)$$

Time complexity if PA1 is run sequentially (single processor) :

$$W(n) = W(n-1) + W(n-2) + O(1) W(n) = O(n^2)$$

Even with parallel spawns, on one processor call all the execution sequential so its same.

1.1.2 (b) Efficienty comparision

PA1 is theoretically work-efficient because it does the same total amount of work as the sequential version O(n^2)

But in practice, it doesnt look like a good strategy because:

- It repeats the same calculations many times.
- Parallel overhead (creating tasks, joining results) makes it slower on a single CPU.
- Even though you can parallelize, it's still exponential work, like using matrix exponentiation

1.1.3 (c) (i) Time complexity of PA1 if you had an infinite number of processors

(ii) Considering this measure, could you elaborate on performance of PA1 when the number of processors increases?

(i) With infinitely many processors, time is determined by the span (critical path length).

The recurrence for span is:

$$S(n) = max(S(n-1), S(n-2) + O(1))S(n) = S(n-1) + O(1)$$

And S(1) = O(1), so we have S(n) = O(n) Answer : O(n)

(ii)Time decreases from $O(n^2)$ to $O(n)$ as processors increase, but beyond a point (when processors exceed the number of ready tasks), no further speedup occurs. Overhead may dominate.

1.1.4 (d) Considering fib_recursive2(n: u32) and PA1, what is the theoretical speedup of the execution if we use P

processors?

Speedup over sequential (O(2^n)):

$$\text{Speedup}(P) \ \leq \ \frac{2^n}{\frac{2^n}{P} + n}$$

For large (n):

$$\text{Speedup}(P) \approx \begin{cases} P, & P \ll \frac{2^n}{n} \\ \frac{2^n}{n}, & P \gg \frac{2^n}{n} \end{cases}$$

Therefore, asymptotically:

$$\text{Speedup}(P) \approx \min\left(P, \frac{2^n}{n}\right)$$

1.2 Exercise 2.

Consider the Linear System Ax = b where

$$A = \begin{bmatrix} 8 & 1 & 1 \\ 1 & 8 & 2 \\ 1 & 2 & 4 \end{bmatrix}$$

and

$$b = \begin{bmatrix} 8 \\ 4 \\ 0 \end{bmatrix}$$

1.2.1 a. Using the initial approximation $x^{(0)} = 0$, carry out 3 interactions of both Jacobi and Gauss-Seidel algorithms to compute $x^{(1)}$,$x^{(2)}$,$x^{(3)}$.

How close are you to the exact solution? Which is a better approximation to the solution - Jacobi or Gauss Seidel?

Given:

$$A = \begin{bmatrix} 8 & 1 & 1 \\ 1 & 8 & 2 \\ 1 & 2 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 8 \\ 4 \\ 0 \end{bmatrix}, \quad x^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(i) Jacobi Method

Jacobi iteration formula:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \Big(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \Big)$$

Iteration 1 ((k=0 →k=1))

$$x_1^{(1)} = \frac{1}{8}(8 - (1 \cdot 0 + 1 \cdot 0)) = 1$$

$$x_2^{(1)} = \frac{1}{8}(4 - (1 \cdot 0 + 2 \cdot 0)) = 0.5$$

$$x_3^{(1)} = \frac{1}{4}(0 - (1 \cdot 0 + 2 \cdot 0)) = 0$$

$$x^{(1)} = \begin{bmatrix} 1 \\ 0.5 \\ 0 \end{bmatrix}$$

Iteration 2 ((k=1 →k=2))

$$x_1^{(2)} = \frac{1}{8}(8 - (1 \cdot 0.5 + 1 \cdot 0)) = 0.9375$$

$$x_2^{(2)} = \frac{1}{8}(4 - (1 \cdot 1 + 2 \cdot 0)) = 0.375$$

$$x_3^{(2)} = \frac{1}{4}(0 - (1 \cdot 1 + 2 \cdot 0.5)) = -0.5$$

$$x^{(2)} = \begin{bmatrix} 0.9375 \\ 0.375 \\ -0.5 \end{bmatrix}$$

Iteration 3 ((k=2 →k=3))

$$x_1^{(3)} = \frac{1}{8}(8 - (1 \cdot 0.375 + 1 \cdot (-0.5))) = 1.015625$$

$$x_2^{(3)} = \frac{1}{8}(4 - (1 \cdot 0.9375 + 2 \cdot (-0.5))) = 0.5078125$$

$$x_3^{(3)} = \frac{1}{4}(0 - (1 \cdot 0.9375 + 2 \cdot 0.375)) = -0.421875$$

$$x^{(3)} = \begin{bmatrix} 1.015625 \\ 0.5078125 \\ -0.421875 \end{bmatrix}$$

(ii) Gauss–Seidel Method

GS iteration formula:

$$x_i^{(k+1)} = \frac{1}{a_{ii}}\Big(b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij}x_j^{(k)}\Big)$$

Iteration 1 ($k = 0 \to k = 1$)

$$x^{(1)} = \begin{bmatrix} 1 \\ 0.375 \\ -0.4375 \end{bmatrix}$$

Iteration 2 ($k = 1 \to k = 2$)

$$x^{(2)} \approx \begin{bmatrix} 1.0078125 \\ 0.483398 \\ -0.493652 \end{bmatrix}$$

Iteration 3 ($k = 2 \to k = 3$)

$$x^{(3)} \approx \begin{bmatrix} 1.001282 \\ 0.498253 \\ -0.499447 \end{bmatrix}$$

Exact Solution Solve (Ax = b):

Third equation:

$$x_1 + 2x_2 + 4x_3 = 0$$

Second equation:

$$x_1 + 8x_2 + 2x_3 = 4$$

First equation:

$$8x_1 + x_2 + x_3 = 8$$

Solving gives exact solution:

$$x^* = \begin{bmatrix} 1 \\ 0.5 \\ -0.5 \end{bmatrix}$$

Comparison after 3 iterations

Jacobi:

$$x^{(3)} = \begin{bmatrix} 1.015625 \\ 0.5078125 \\ -0.421875 \end{bmatrix}$$

Error:

$$e^{(3)} = x^{(3)} - x^* = \begin{bmatrix} 0.015625 \\ 0.0078125 \\ 0.078125 \end{bmatrix}$$

Gauss–Seidel:

$$x^{(3)} \approx \begin{bmatrix} 1.001282 \\ 0.498253 \\ -0.499447 \end{bmatrix}$$

Error:

$$e^{(3)} = x^{(3)} - x^* \approx \begin{bmatrix} 0.001282 \\ -0.001747 \\ 0.000553 \end{bmatrix}$$

Gauss–Seidel is much closer to the exact solution after the same number of iterations.

Final Answer

- Jacobi $x^{(3)} = [1.015625, 0.5078125, -0.421875]$
- Gauss–Seidel $x^{(3)} \approx [1.001282, 0.498253, -0.499447]$
- Exact solution $x^* = [1, 0.5, -0.5]$

Conclusion: Gauss–Seidel gives a better approximation than Jacobi after 3 iterations.

1.2.2 (b) Write the interactions of each method in the form $ x^{(k+1)} = Tx^{(k)}+c$. Find matrices T and vectors c of explicitly.

Decompose (A = D + L + U):

$$D = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \quad L = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 2 & 0 \end{bmatrix}, \quad U = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Jacobi Method

Iteration formula:

$$x^{(k+1)} = D^{-1}(b - (L+U)x^{(k)})$$

$$T_J = -D^{-1}(L+U), \quad c_J = D^{-1}b$$

Compute (D^{-1}):

$$D^{-1} = \begin{bmatrix} \frac{1}{8} & 0 & 0 \\ 0 & \frac{1}{8} & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix}$$

Compute (L+U):

$$L + U = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$$

Multiply:

$$D^{-1}(L+U) = \begin{bmatrix} 0 & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{8} & 0 & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & 0 \end{bmatrix}$$

$$T_J = -D^{-1}(L+U) = \begin{bmatrix} 0 & -0.125 & -0.125 \\ -0.125 & 0 & -0.25 \\ -0.25 & -0.5 & 0 \end{bmatrix}$$

Compute (c_J):

$$c_J = D^{-1}b = \begin{bmatrix} 1 \\ 0.5 \\ 0 \end{bmatrix}$$

Gauss-Seidel Method

Iteration formula:

$$x^{(k+1)} = (D+L)^{-1}(b - Ux^{(k)})$$

$$T_{GS} = -(D+L)^{-1}U, \quad c_{GS} = (D+L)^{-1}b$$

Compute (D+L):

$$D+L = \begin{bmatrix} 8 & 0 & 0 \\ 1 & 8 & 0 \\ 1 & 2 & 4 \end{bmatrix}$$

Inverse ((D+L)^{-1}):

$$(D+L)^{-1} = \begin{bmatrix} 0.125 & 0 & -0.0625 \\ -0.015625 & 0.125 & 0.0078125 \\ -0.0234375 & -0.0625 & 0.25 \end{bmatrix}$$

Compute $T_{GS} = -(D+L)^{-1}U$, where

$$U = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{GS} = \begin{bmatrix} 0 & -0.125 & -0.125 \\ 0 & 0.015625 & -0.234375 \\ 0 & 0.0234375 & 0.1484375 \end{bmatrix}$$

Compute $c_{GS} = (D+L)^{-1}b$:

$$c_{GS} = \begin{bmatrix} 1 \\ 0.375 \\ -0.4375 \end{bmatrix}$$

1.2.3 (c) Compare the spectral radius of the matrices T in question (b).

Based on the spectral radius values, what can you say about the convergence of the m

$$T_J = \begin{bmatrix} 0 & -0.125 & -0.125 \\ -0.125 & 0 & -0.25 \\ -0.25 & -0.5 & 0 \end{bmatrix}, \quad T_{GS} = \begin{bmatrix} 0 & -0.125 & -0.125 \\ 0 & 0.015625 & -0.234375 \\ 0 & 0.0234375 & 0.1484375 \end{bmatrix}$$

Jacobi T matrix eigenvalues

Characteristic polynomial:

$$p(\lambda) = \det(T_J - \lambda I) = \det \begin{bmatrix} -\lambda & -0.125 & -0.125 \\ -0.125 & -\lambda & -0.25 \\ -0.25 & -0.5 & -\lambda \end{bmatrix} = 0$$

Compute determinant step by step:

$p(\lambda) = -\lambda[(-\lambda)(-\lambda)-(-0.25)(-0.5)]-(-0.125)[(-0.125)(-\lambda)-(-0.25)(-0.5)]+(-0.125)[(-0.125)(-0.5)-(-\lambda)($

Simplify:

$$p(\lambda) = -\lambda^3 + 0.171875\lambda - 0.0234375 = 0$$

Multiply by (-1):

$$\lambda^3 - 0.171875\lambda + 0.0234375 = 0$$

Numerical roots:

$$\lambda_1 \approx 0.31, \quad \lambda_2 \approx 0.1609, \quad \lambda_3 \approx -0.4709$$

Spectral radius:

$$\rho(T_J) = \max |\lambda_i| \approx 0.4709$$

Gauss-Seidel T matrix eigenvalues

$$T_{GS} = \begin{bmatrix} 0 & -0.125 & -0.125 \\ 0 & 0.015625 & -0.234375 \\ 0 & 0.0234375 & 0.1484375 \end{bmatrix}$$

Notice first column all zeros → one eigenvalue (= 0).

Let

$$B = \begin{bmatrix} 0.015625 & -0.234375 \\ 0.0234375 & 0.1484375 \end{bmatrix}$$

Eigenvalues of (B):

$$\lambda^2 - \mathrm{tr}(B)\lambda + \det(B) = 0$$

$\mathrm{tr}(B) = 0.015625+0.1484375 = 0.1640625, \quad \det(B) = 0.015625 \cdot 0.1484375-(-0.234375) \cdot 0.0234375 = 0.0078125$

$$\lambda^2 - 0.1640625\lambda + 0.0078125 = 0$$

Complex roots:

$$\lambda = 0.08203125 \pm 0.032915i, \quad |\lambda| \approx 0.08838$$

Spectral radius:

$$\rho(T_{GS}) \approx 0.08838$$

Comparison

$$\rho(T_J) \approx 0.471, \quad \rho(T_{GS}) \approx 0.088$$

Both methods converge ((< 1)), but **Gauss-Seidel converges much faster** because its spectral radius is smaller.

Convergence speed ratio:

$$\frac{\rho(T_J)}{\rho(T_{GS})} \approx \frac{0.471}{0.088} \approx 5.35$$

So Gauss-Seidel converges roughly 5.35 times faster per iteration than Jacobi.

Final Answer:

$$\rho(T_J) \approx 0.47, \quad \rho(T_{GS}) \approx 0.088$$

Both converge, but Gauss-Seidel is significantly faster due to smaller spectral radius.

1.2.4 d. For the faster method, how many iterations you would expect are needed to obtain five decimal place accuracy?

For Gauss-Seidel with $\|T\|_\infty = 0.25$:

$$\|T\|_\infty^k \times \|x^{(0)} - x^*\|_\infty \leq 10^{-5}$$
$$0.25^k \times 1 \leq 10^{-5}$$
$$0.25^k \leq 10^{-5}$$
$$k \log_{10}(0.25) \leq -5$$
$$-0.60206\,k \leq -5$$
$$k \geq \frac{5}{0.60206} \approx 8.30$$

So:

$$k = 9 \text{ iterations}$$

Check with iteration count:

$$0.25^9 \approx 3.81 \times 10^{-6} < 10^{-5}$$
$$0.25^8 \approx 1.53 \times 10^{-5} > 10^{-5}$$

Therefore, **9 iterations** are needed for guaranteed 5 decimal place accuracy using the norm bound.

Answer:
9 iterations

1.3 Exercise 3.

Let b be strictly diagonally dominant: \$ |a_{ii}| > *{j i}* /*A*{ij}|\$ Show that both Jacobi and Gauss-Seidel methods converges.

Given: A is strictly diagonally dominant:

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}| \quad \text{for all } i$$

Jacobi method convergence

Jacobi iteration matrix: $T_J = -D^{-1}(L + U)$
Let $T_J = [t_{ij}]$.

For $i \neq j$: $t_{ij} = -\frac{a_{ij}}{a_{ii}}$
For $i = j$: $t_{ii} = 0$

Row sum of absolute values for row i:

$$\sum_{j=1}^{n} |t_{ij}| = \sum_{j \neq i} \frac{|a_{ij}|}{|a_{ii}|} = \frac{1}{|a_{ii}|} \sum_{j \neq i} |a_{ij}|$$

From diagonal dominance:

$$\sum_{j \neq i} |a_{ij}| < |a_{ii}| \quad \Rightarrow \quad \sum_{j=1}^{n} |t_{ij}| < 1 \quad \text{for all } i$$

So:

$$\|T_J\|_\infty = \max_i \sum_j |t_{ij}| < 1$$

Since $\rho(T_J) \leq \|T_J\|_\infty < 1$, Jacobi converges.

Gauss-Seidel method convergence

GS iteration: $x^{(k+1)} = (D + L)^{-1}(b - Ux^{(k)})$
Let $T_{GS} = -(D + L)^{-1}U$.

Let λ be an eigenvalue of T_{GS} with eigenvector $x \neq 0$:

$$T_{GS}x = \lambda x \quad \Rightarrow \quad -(D + L)^{-1}Ux = \lambda x$$

Multiply both sides by $-(D+L)$:

$$Ux = -\lambda(D+L)x \quad \Rightarrow \quad \lambda(D+L)x + Ux = 0$$

$$\lambda Dx + \lambda Lx + Ux = 0$$

Component-wise for $i = 1, \ldots, n$:

$$\lambda a_{ii}x_i + \lambda \sum_{j=1}^{i-1} a_{ij}x_j + \sum_{j=i+1}^{n} a_{ij}x_j = 0$$

$$\lambda a_{ii}x_i = -\lambda \sum_{j=1}^{i-1} a_{ij}x_j - \sum_{j=i+1}^{n} a_{ij}x_j$$

Let $r = \max_i |x_i|$, and p be the index where $|x_p| = r > 0$. Then for $i = p$:

$$|\lambda a_{pp}x_p| \le |\lambda| \sum_{j=1}^{p-1} |a_{pj}||x_j| + \sum_{j=p+1}^{n} |a_{pj}||x_j|$$

Since $|x_j| \le r$:

$$|\lambda||a_{pp}|r \le |\lambda| \sum_{j=1}^{p-1} |a_{pj}|r + \sum_{j=p+1}^{n} |a_{pj}|r$$

Divide by $r > 0$:

$$|\lambda||a_{pp}| \le |\lambda| \sum_{j=1}^{p-1} |a_{pj}| + \sum_{j=p+1}^{n} |a_{pj}|$$

Rearrange:

$$|\lambda| \left(|a_{pp}| - \sum_{j=1}^{p-1} |a_{pj}| \right) \le \sum_{j=p+1}^{n} |a_{pj}|$$

From strict diagonal dominance:

$$|a_{pp}| > \sum_{j \ne p} |a_{pj}| = \sum_{j=1}^{p-1} |a_{pj}| + \sum_{j=p+1}^{n} |a_{pj}|$$

Let $S_1 = \sum_{j=1}^{p-1} |a_{pj}|$, $S_2 = \sum_{j=p+1}^{n} |a_{pj}|$:

$$|\lambda| \leq \frac{S_2}{|a_{pp}| - S_1} < 1$$

Since λ was arbitrary, $\rho(T_{GS}) < 1$, so Gauss-Seidel converges.

Conclusion: For strictly diagonally dominant matrices, both Jacobi and Gauss-Seidel methods converge.